

svpview

Valeport SWiFT acquisition, display and winch depth reporting — in C99 and SDL2, with the design reasoning

Application	svpview 2026.08.12
Replaces	Valeport Ocean + VigoDepthRelay, in this workflow
Instruments	SWiFT SVP, SWiFT CTD, SWiFT <i>plus</i>
Language	C99 — 8 441 lines, one dependency (SDL2)
Platforms	Linux, macOS, Windows (Win32 layer outstanding)
Repository	C-MAX/src/svpview
Date	12 August 2026

Engineering working document — simulator-proven, not yet hardware-proven

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1 What it is

svpview configures a Valeport SWiFT sound-velocity or CTD profiler, downloads the casts it recorded, derives and plots them, exports to the formats the survey software downstream wants, and reports each cast's achieved depth to a C-MAX Vigo winch. It is one statically-laid-out binary with no installer, no runtime, and no configuration beyond a settings file it writes itself.

1.1 Scope, and the line it does not cross

It does **not** control the winch. Casts are commanded from the winch's own panel and web UI, which is where the operator already is and where the safety interlocks live. `svpview` only *reports* the depth a cast reached, over the profiler UDP protocol Vigo already listens for on port 8090, so it drops into an existing workflow in place of `VigoDepthRelay` without becoming a second thing that can drive a winch.

WORTH KNOWING

Vigo also exposes a complete ASCII control API on TCP 8092. It is documented in `docs/VIGO_INTERFACE.md` as *available and deliberately unused*, so the decision is recorded rather than rediscovered and re-argued in a year.

1.2 The one moment it makes the winch move

There is a single exception, and it comes from the profiler protocol rather than any control API: if the file transfer fails — typically a Bluetooth link dropping with the instrument hanging at the transfer point — `svpview` sends `F`. Vigo jogs the spool in by 100 mm to shorten the Bluetooth path, re-checks its proximity switch and answers `BTRST`, and the download is retried. Bounded retry count, then it stops and says so.

2 Design intent

Seven rules. They are not aspirations written after the fact — each one has code and test consequences that are pointed out where they appear later in this document.

2.1 Never present a number the data does not support

Every value on screen can be traced to a measurement or is marked as absent. A cast logged without a GPS fix does not get plotted at 0°N 0°E — it is counted in the chart's panel as “not shown: logged with no GPS fix”. A status sentence whose checksum fails is parsed, flagged and counted, not silently accepted and not silently dropped. A depth the winch will reject is explained *before* the datagram goes out, in the operator's terms.

2.2 Treat every length in a file as hostile

The `.bin` reader takes its header size from a field inside the file. That field can point past the end of the file, and a corrupt one did produce a genuine out-of-bounds read during development — found by the test that deliberately corrupts it, under AddressSanitizer. The reader now walks a bounded cursor that latches on the first overrun and returns zeroes thereafter, so a truncated or garbled file yields a short cast or a clear refusal, never a crash.

2.3 Bound the work a frame can do

A plot must not cost more to draw because a cast holds more samples than the plot has pixels. Each trace is decimated to the pixel grid as it is drawn, so a 1200-sample cast costs what a few hundred segments cost and looks identical.

This one arrived the hard way, and it is the best evidence in this document that the rule matters: overlaying four casts with all four traces — roughly 9600 line segments, each of them a quad — crossed Nuklear's 16-bit vertex index limit of 65535 and the assertion inside its draw list *killed the program*. Ticking four checkboxes with four casts open is not an edge case. The ceiling is now 32-bit (which needed a fix to the vendored SDL backend, where the index size is hard-coded to two bytes — an upstream bug), and the decimation keeps a frame's vertex count tied to the size of the plot rather than to the data.

2.4 The interface must never be blocked by I/O

Every link is non-blocking and polled once per frame. There is no operation that stops the window repainting, which is why there is no need for a progress dialog that lies. See §9.

2.5 One layout system, no special cases

Pages are built from the same row helpers, so labels and fields share edges across every page. Dialogs all carry Cancel / Apply / OK in that order, right aligned, with the body scrolling under a pinned button row and a title-bar close; each is measured as it draws and sized to its own content, so a dialog never hides its controls behind a scrollbar and the adapter picker is exactly one row taller when a second adapter appears.

2.6 Scale with the display, both ways

Every metric is written at scale 1 and multiplied by the UI scale as it draws; the font atlas is baked at physical pixel size so text is crisp rather than magnified. The window opens at the design size *times* that scale, clamped to the desktop's usable bounds — on a 192 dpi panel that is 2560×1640 instead of a postage stamp.

2.7 Say what is not known

Where the integration guide is silent or our copy of it is defective, the gap is documented as a gap. The acknowledged-extraction command [#433](#) is unimplemented because the protocol figure is blank in the PDF; it is listed as deferred with the reason, not quietly omitted.

THE SHORT VERSION

A survey day costs more than the software. The program is allowed to refuse, to explain and to be slow; it is not allowed to crash, to lose a cast, or to state a depth it cannot justify.

3 The screen

A status strip along the top that is always true, a navigation rail, the page, and an action bar that reports what the program is doing and what the winch last accepted. Six pages.

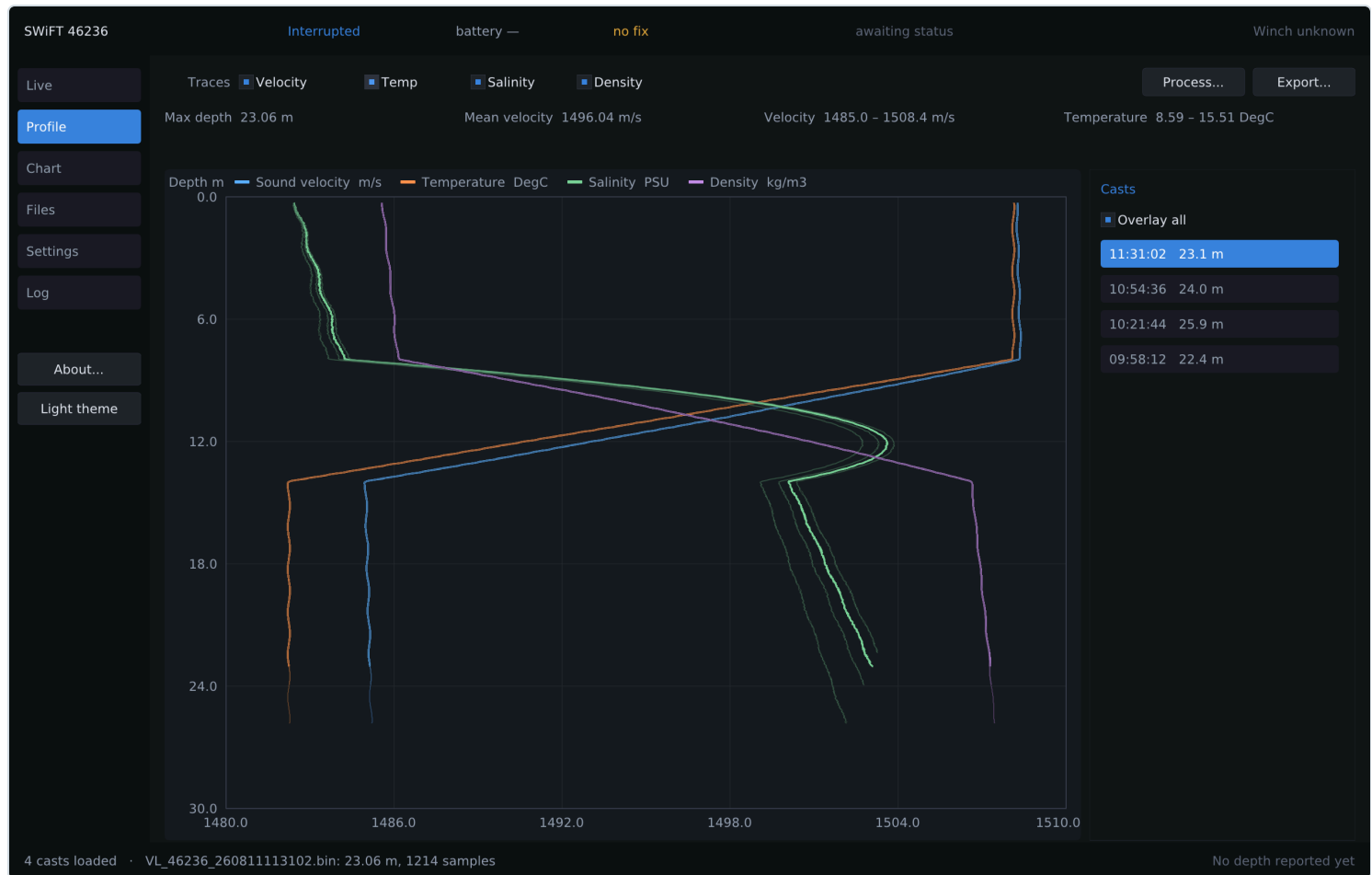


Figure 1. The **Profile** page: four casts overlaid with all four traces on — the case that used to abort the program (§2.3). Each trace has its own scale so it uses the full width; the selected cast is opaque and the rest are dimmed, which is what makes an overlay readable. The summary line under the controls belongs to the selected cast. The status strip names the instrument, the link state, battery hours, the last fix, the deploy verdict and whether the winch is answering.

3.1 Status strip — the six things that matter

Instrument and serial, link state, battery hours remaining, last position, the ready-to-deploy verdict, and the winch. The deploy verdict is not a flag repeated back: when the instrument says it is not ready, the strip and the Live page explain *why* in the operator's terms — no fix, or the trigger depth is not set — rather than showing a code to look up. The minimum window width is set by this strip, because "NOT ready to deploy" truncating mid-word is the one caption that must never be misread.

3.2 Live

Continuous-mode time series of the primary observable, with the instrument's identity, the last recorded file name and a running count of checksum failures — a slow rise there is the first sign of a marginal Bluetooth link.

3.3 Profile

Sound velocity, temperature, salinity and density against depth. Multi-cast overlay, cursor readout of the nearest sample, axis ranges snapped to round numbers. Tick labels are dropped where they would collide but every gridline is kept — at UI scale 2 on a narrow plot there is not room for all six numbers, and overlapping digits are worse than fewer of them.

3.4 Chart

Where the casts were taken. §4.

3.5 Files

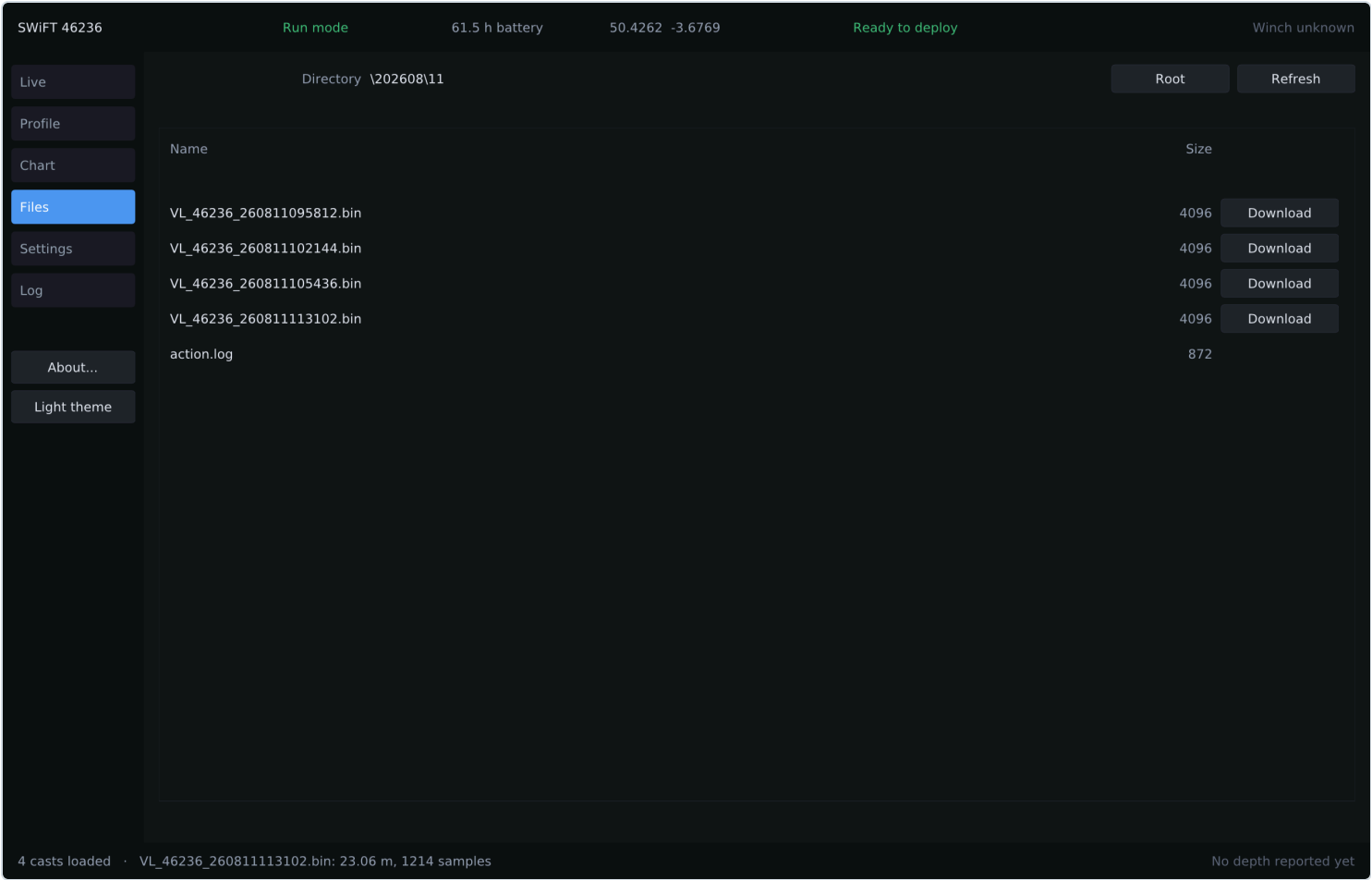


Figure 2. The **Files** page browsing the instrument's card. Directory navigation is by #400/#406; each file has its own Download button, and the download streams to a temporary file which is parsed and only then kept. Nothing is deleted from the card except on an explicit operator action.

3.6 Settings

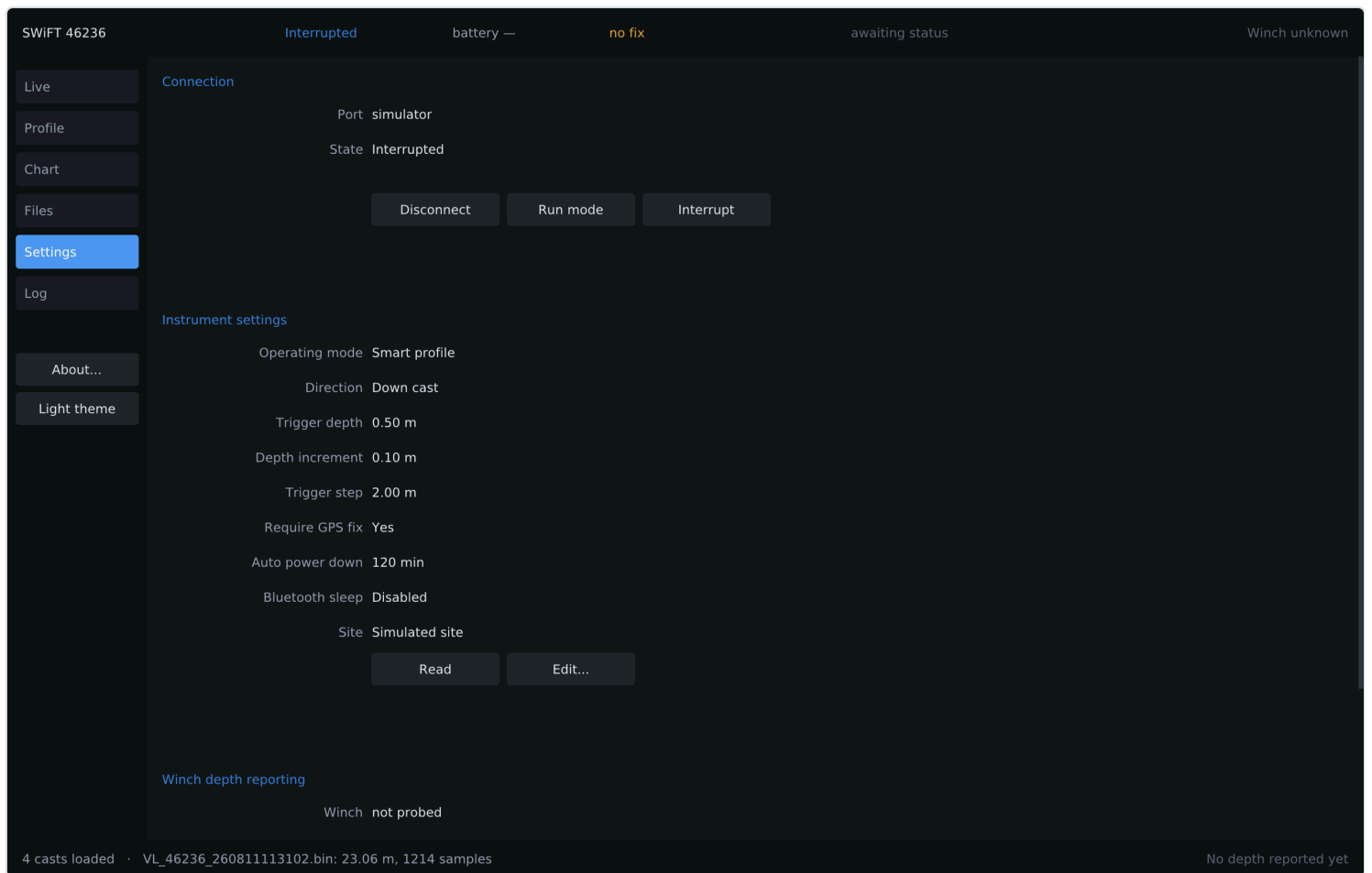


Figure 3. Settings, showing the configuration as last read from the instrument. Writes send only what changed and read it back to verify; the network adapter picker (§8) lives here too.

3.7 Log

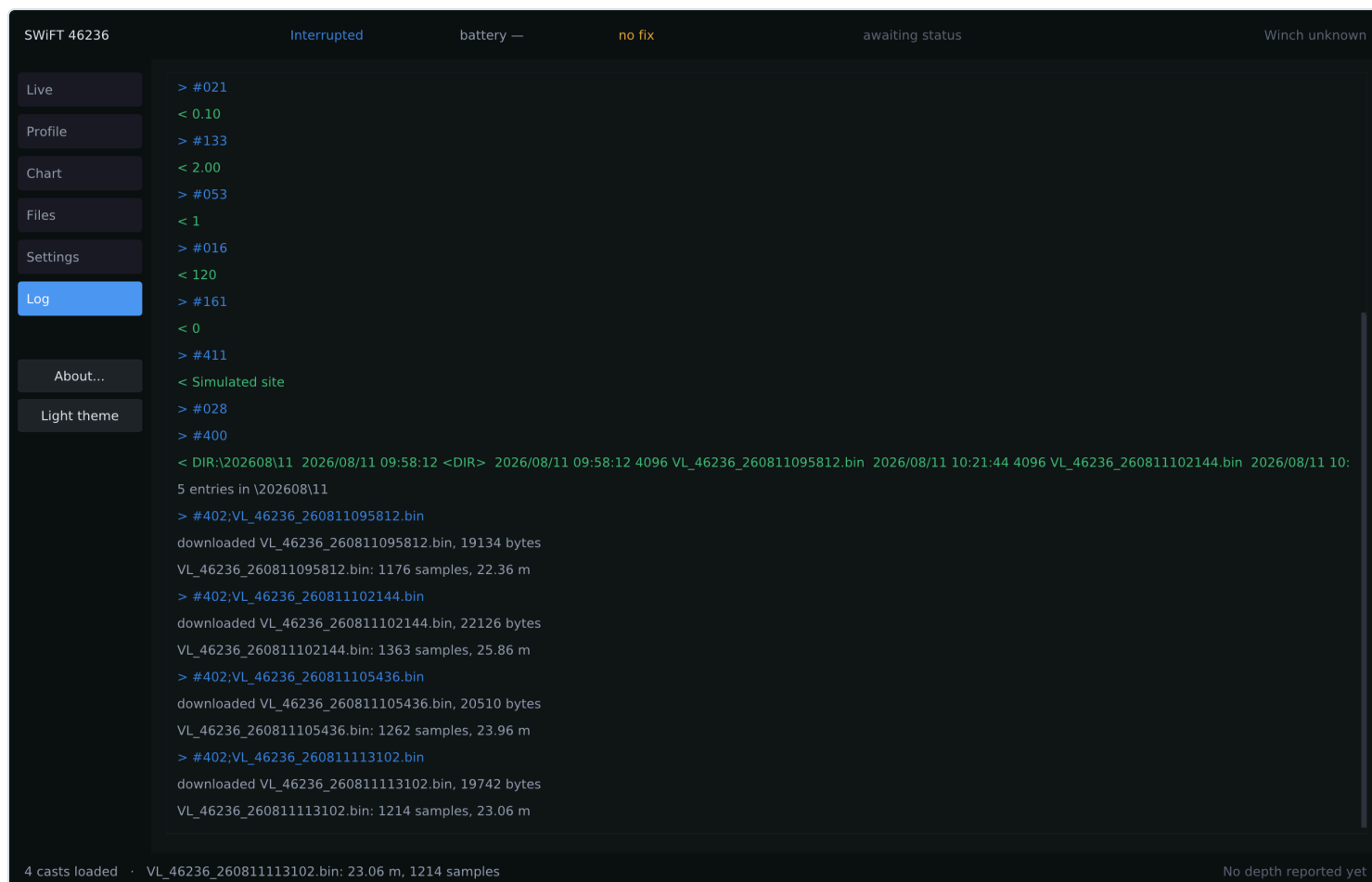


Figure 4. The **Log** — one line per protocol exchange, with control characters sanitised so a multi-line instrument reply does not draw as boxes. It follows the newest line, but stops the moment the operator scrolls up and resumes when they scroll back to the bottom: having the view yanked back mid-read makes the page useless during a download, which is exactly when it is needed.

3.8 Themes, and knowing which build you are looking at

Dark and light come from one role-based palette table, so a widget never hard-codes an RGB value and the two themes cannot drift apart. Same layout, same metrics, one word changed on the button that switches them.

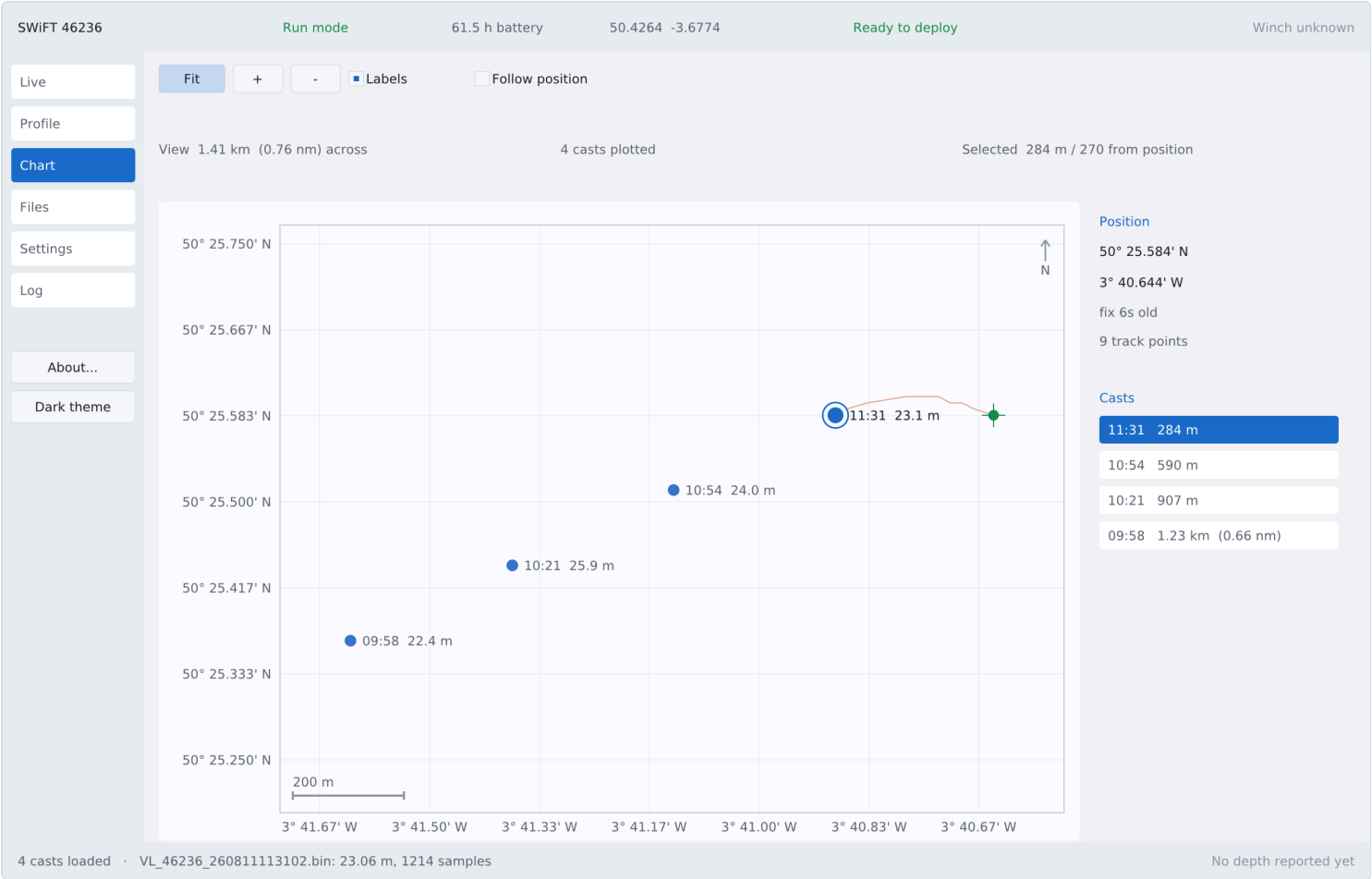


Figure 5. The light theme, on the same page as Figure 7. Nothing about the layout changes — only the palette, addressed by role: `panel`, `plot_grid`, `text_dim`, `accent`. A deck in sunlight and a dark bridge at 3 a.m. are different rooms.

The window title carries the version and, once connected, the instrument and port — so a screenshot identifies the build, and two profilers open at once can be told apart from the taskbar:

```
svpview 2026.08.12 - SWiFT 46236 on /dev/ttyUSB0
```

The About box adds the build stamp, SDL at runtime *and* the version it was compiled against (those differ on a machine with a stale library), the UI scale in force, and the connected instrument: the questions a support call opens with, answerable without asking the operator to run anything.



Figure 6. About — one OK button, because Apply and Cancel have nothing to act on in a dialog that changes nothing.

4 The chart

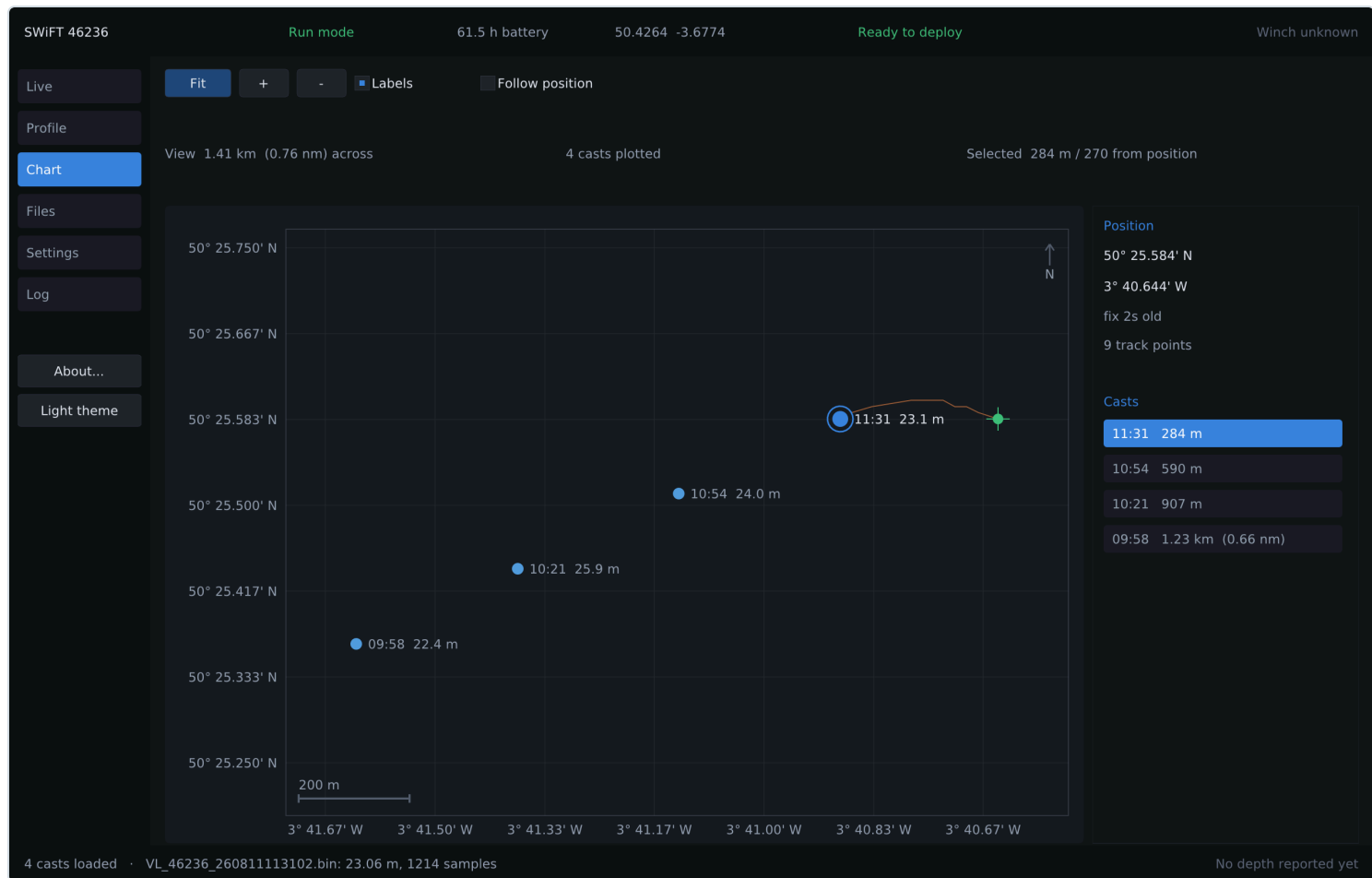


Figure 7. The **Chart** page. Cast markers with their time and maximum depth, the track the instrument has been broadcasting, a graticule on whole minutes and seconds, a scale bar and a north arrow. The panel gives the live position, the track point count, and every cast's range from where the vessel is now. Drag pans, the wheel zooms about the cursor, a click selects the cast under it — and the Profile page follows that selection.

4.1 A chart, not a map

There is no basemap imagery, and that is a decision rather than an omission. Tiles would mean a tile server, an image decoder, a cache, a licence and an internet connection on a boat that has none. What a positioning plot has to do — show the geometry between the casts, the track and the vessel, to a stated scale — needs none of that. Nothing was added to the dependency list to build this page.

4.2 A tangent plane, on purpose

Positions are projected onto a local tangent plane about the view centre, using the WGS84 metres-per-degree series at the working latitude. For casts within one survey area — tens of kilometres at most — that is accurate to centimetres, and it is *better* than a spherical formula on a mean radius, which is out by roughly 0.3%: 3 m in every kilometre. Distance and bearing fall back to the spherical forms once two positions are more than a third of a degree apart, so a stray fix on the other side of the world still gives a sane answer instead of nonsense.

The forward projection and the distance function are checked against each other in the test suite, so the chart cannot draw one separation while the readout states another.

4.3 The graticule is sexagesimal

Gridlines land on 30°, 10°, 5°, 2°, 1°, then whole minutes, then whole seconds, then tenths — never on a round number of metres. Labels are degrees and decimal minutes, the form written in a survey log, and carry exactly the resolution the graticule is drawn at and no more. The scale bar is separately a round 1 / 2 / 5 distance, because a scale bar has to be divisible by eye.

4.4 Two position sources, two precisions, stated

Source	Carries	Resolution	Used for
Cast file header	float32 lat, lon	sub-metre	cast markers
\$PVBB broadcast	four decimal places	~11 m of latitude	live marker, track

This is why the track keeps only fixes more than 15 m from the last one. Below that figure a *stationary* instrument whose GPS wanders across a quantisation cell reports two positions 11 m apart for ever, and the track would fill with jitter that looks like movement. The threshold is a property of the broadcast, not a guess, and the simulator reproduces the four-decimal quantisation rather than hiding it.

5 Talking to the instrument

5.1 The session

230 400 8N1 over Bluetooth or USB. # interrupts run mode; the instrument answers with a > prompt and accepts #NNN commands; #028 returns it to run mode. The connect sequence is interrupt → identify → read every setting → back to run mode, so the program never sits at a prompt it forgot to leave, and every step carries a deadline: a pulled cable leaves the link closed with a reason, not hung.

5.2 Four findings that cost real time

Each of these is now a regression test.

Finding	What went wrong, and what it means
The prompt is a > at the start of a line	Directory listings contain <DIR>. Treating any > as the prompt truncated every listing at the first subdirectory.
The command echo precedes the file data	On #402 the instrument echoes the command before the payload. Left in, it shifts the header by a few bytes and every download parses as corrupt. Echo skipped, trailing prompt trimmed.
Basic extraction has no terminator	End of file is an idle timeout — 900 ms of silence. Worth knowing before designing a progress indicator around a length that never arrives.
The header size counts itself through the ETX	Confirmed rather than assumed: walking the documented fields gives 246 / 295 / 312 bytes for the three firmware variants, matching the totals the guide quotes independently. Asserted in <code>tests/test_binfile.c</code> .

5.3 The checksum, and why bad ones are kept

The \$PVBB checksum is a plain XOR of everything between \$ and *. Our copy of the guide does not reproduce its own worked examples — they are out by a constant 0x70, so characters were lost in the document. The algorithm is nevertheless confirmed, because the *deltas* between successive examples reproduce exactly. A sentence with a failing checksum is therefore parsed, flagged and counted rather than discarded: on a link where the guide itself is lossy, throwing away data on a checksum we cannot fully verify against the vendor’s own text would lose real casts.

6 Oceanography

All of it in `sv_ocean.c` — portable, no I/O, unit-tested against published values. An SVP measures sound speed and salinity is inverted from it; a CTD measures conductivity and sound speed is computed. Either way all five derived columns end up populated, which is what the export writers and the plot expect, so nothing downstream needs to know which instrument produced the cast.

Quantity	Method	Check value
Depth from pressure	UNESCO 1983, with the gravity term at the cast's own latitude	9712.653 m at 10000 dBar, 30°N
Sound speed	Chen & Millero	$c(35, 0, 0) = 1449.14$ m/s
Salinity	PSS-78	round-trips with conductivity
Density	EOS-80	$\rho(35, 5, 0) = 1027.67547$ kg/m ³
Salinity from sound speed	numeric inverse of Chen & Millero	inverts to 1e-6
Conductivity from salinity	numeric inverse of PSS-78	inverts to 1e-6

WHY THE LATITUDE MATTERS

Pressure-to-depth carries a gravity term. Using a nominal latitude instead of the cast's own is a systematic depth error that varies with where you are working — small, but free to get right, since the cast already knows where it was taken.

7 Processing and export

7.1 Processing

Applied to the selected cast, each step either fills in derived columns or removes samples — none of them ever grow the array, so the single allocation made by the file reader is the only one a cast ever needs.

- **Down/up-cast split.** The turning point is the deepest sample; a minimum depth removes the surface soak and the recovery tail.
- **Despike.** A sample whose sound speed differs from the median of its window by more than a tolerance goes; the count removed is reported rather than applied silently.
- **Thin.** Largest-triangle decimation on depth against sound speed — it drops the samples that contribute least to the *shape*, so a 1262-sample cast thinned to 100 still has its thermocline.
- **Depth bin.** Fixed bins, one entry per occupied bin at the bin's mean depth.

7.2 Export

CSV, Valeport VP2, Kongsberg `.asvp`, Caris `.svp`, Hypack `.vel`.

Every file is written to a temporary file and renamed into place. A survey machine that loses power mid-write then has either the old file or the new one — never a half-written sound velocity profile that the processing software will happily load and quietly misuse.

8 The winch link

One UDP socket on port 8090 and three message types. Vigo dispatches on character 0 only, which is why the message shapes below are what they are.

Msg	Sent	Reply
Q	at startup and on a timer, to prove the winch is there	VIGO responding
V	VP-<NNN>,Valeport-Winch-Go,<depth>	ACK
F	download failed — ask for the spool to be jogged in	BTRST

8.1 Five ways to be ignored

Vigo silently discards a depth report under five conditions, and all five look identical from outside: “the winch didn’t see it”. No cast performed yet this session; the message is byte-identical to the previous one; the cast was aborted; the depth is not finite or not positive; or the depth is outside 0.1×–10× the requested cast depth. svpview mirrors all five rules locally and explains the rejection *before* transmitting, so the failure appears as a sentence on the Log page rather than as a dive table that mysteriously has no entry.

The dedupe rule is why the sequence number increments per cast: two consecutive casts to the same depth would otherwise produce identical datagrams and the second would be dropped. This was verified end to end against a stand-in for Vigo’s listener — the same depth sent again as VP-002 was acknowledged, where a byte-identical repeat is not.

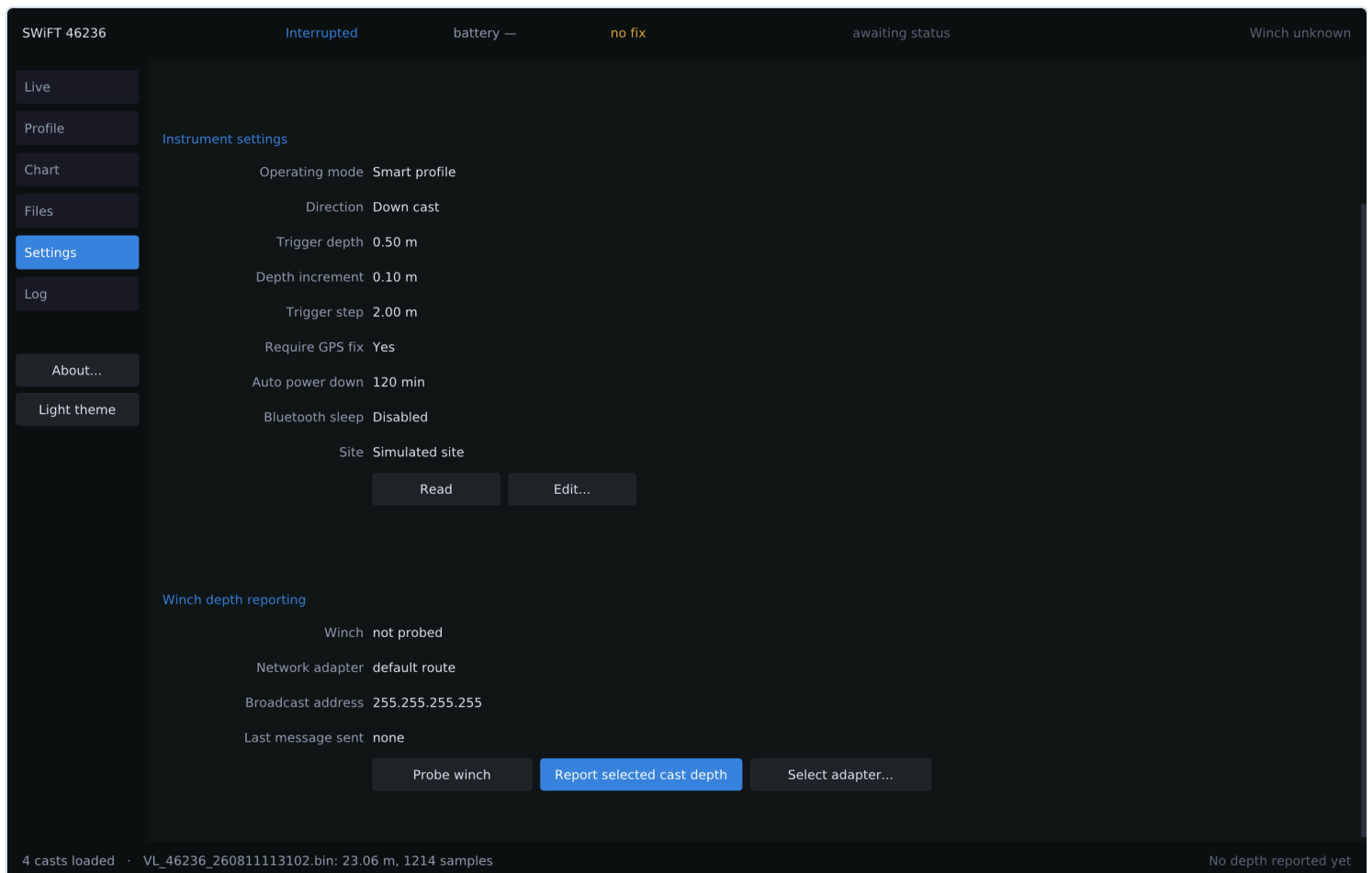


Figure 8. The winch section of the **Settings** page: which adapter is in use, the broadcast address that follows from it, what was last sent, and the three actions. “Report selected cast depth” is the only button in the program that sends a depth, and it names what it will send.

8.2 Which adapter, and the broadcast address

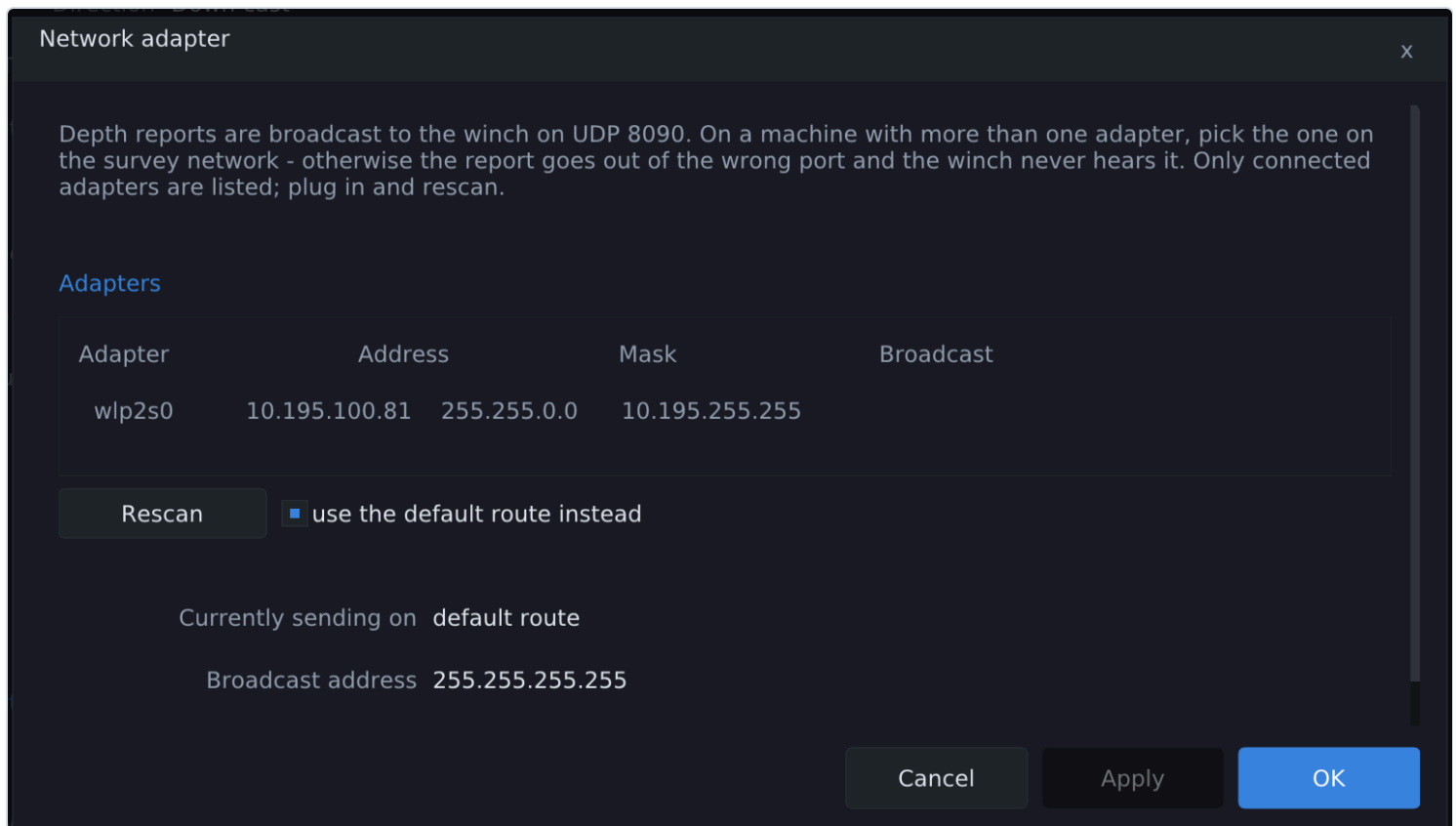


Figure 9. The adapter picker lists only *connected* broadcast-capable adapters — up, running and holding an address — with the address, mask and the broadcast that will actually be used, and binds the winch socket to the chosen one.

The directed broadcast is computed as $(ip \ \& \ mask) \ | \ \sim mask$ rather than taken from the operating system: the Windows path has to compute it anyway, and a wrong broadcast address sends the depth report to the wrong subnet, where nothing complains and nothing arrives. Verified against a real /28 adapter that appeared mid-test — 172.20.10.2 / 255.255.255.240 → 172.20.10.15.

9 Architecture

9.1 Layers

Four, and the dependencies only point one way. The core knows nothing about SDL and does no I/O, which is what makes it testable; everything OS-specific is behind one header.

Layer	Modules	Knows about
UI	<code>sv_ui</code> , <code>sv_plot</code> , <code>sv_chart</code> , <code>sv_theme</code>	SDL2, Nuklear, the app
App	<code>sv_app</code> , <code>sv_sim</code>	the core and the platform layer
Core	<code>sv_ocean</code> , <code>sv_geo</code> , <code>sv_proto</code> , <code>sv_binfile</code> , <code>sv_profile</code> , <code>sv_export</code> , <code>sv_vigo</code> , <code>sv_config</code>	nothing but C and libm
Platform	<code>plat_posix</code> , <code>plat_win32</code>	the OS, behind <code>plat.h</code>

9.2 Why there are no threads

The architecture note originally sketched one I/O thread per link. That was replaced before it was written, and the reasoning is recorded in the file rather than lost: at 230 400 baud a frame's worth of serial data is under 400 bytes. A thread would buy nothing measurable and cost a class of race conditions this program cannot afford. Everything is non-blocking and polled once per frame from one place.

```
/* the whole concurrency design */
while (!quit) {
    pump_sdl_events();
    sv_app_poll(app);      // serial, UDP, simulator - none of them block
    sv_ui_frame(ui, w, h);
    render();
}
```

No locks anywhere in the program. A bug can therefore be reasoned about by reading one path.

A BUILD-SYSTEM BUG WORTH REPEATING

The Makefile originally had no header dependency tracking. Adding a field to a struct in `plat.h` rebuilt some objects and not others — which does not fail to link, it silently reads the wrong fields. It surfaced as the winch page showing a subnet mask where the broadcast address should be. Now built with `-MMD -MP` and the generated `.d` files included. Any C project without this has the same bug waiting.

10 How it is tested

10.1 399 assertions, always sanitised

`make test` compiles every suite with AddressSanitizer and UndefinedBehaviorSanitizer. A test that only passes without them has not passed.

Suite	Checks	Covers
<code>test_geo</code>	106	metres per degree, distance, bearing, projection round-trip, graticule steps, position formatting
<code>test_proto</code>	85	command codec, every documented sentence, framing, directory listings, the prompt rule
<code>test_profile</code>	71	derive, cast split, despike, thinning, binning
<code>test_binfile</code>	68	three header variants, SV and CTD, ASCII and UTF-16 site info, corrupt-length cases
<code>test_vigo</code>	38	message building, reply classification, the five silent-drop rules
<code>test_ocean</code>	31	depth, sound speed, salinity, density and both inverses

10.2 The simulator is the important part

`sv_sim.c` emulates a SWiFT *at the wire*: command echo, the `>` prompt, `$PVBB` broadcasts with their real four-decimal quantisation, and a synthetic 25 m cast — warm mixed layer, thermocline, cold isothermal layer, sampled at 32 Hz down and back up — delivered as a real `.bin` over `#402`. The session code, the parsers and the download path are all exercised, not bypassed. The simulated vessel is under way at 4 knots on a gently turning line, so successive casts land at different positions and the chart has real geometry to draw.

The emulator deliberately does *not* share arithmetic with the code under test — it computes its own sound speed and its own vessel motion from constants written out locally. Sharing a helper would let a test pass because both sides made the same mistake.

10.3 What the tests did not catch

Worth stating, since it is the honest half of the picture. The vertex-limit abort in §2.3 is not reachable from any unit test: it needs a real Nuklear draw list, a real font atlas and a window of a particular size. It was found by driving the built application through an ordinary sequence — open four casts, tick the trace boxes — and watching it die. 399 green assertions say nothing about that class of defect.

The screenshots in this document are therefore not decoration. Every one was taken by scripting the real binary at the real UI scale, and **three** defects fell out of doing it: the abort above, a summary row that was never drawn because it landed a few pixels beyond the panel’s clip, and a scroll offset shared between pages — scroll the settings page down, switch to the chart, and the chart’s controls were off the top of a page with no scrollbar to bring them back. All three are invisible to a test suite and obvious in a screenshot.

WHERE A TEST WAS WRONG AND THE CODE WAS RIGHT

A test asserting that a corrupt header size makes a file unreadable failed. The code was correct: a bad size field is legitimately recovered by walking the documented field layout. The test was fixed to expect recovery, and a second case added where the ETX is destroyed too, which genuinely is unrecoverable. Reading the failure before believing it is part of the method.

11 What is proven and what is not

Everything in this document runs. Most of it has only ever run against the simulator and the unit tests. That distinction is the most important sentence in the document, so it gets a table rather than a footnote.

Area	State	Meaning
Protocol parsers, oceanography, geodesy, processing, export	proven	399 assertions under ASan and UBSan, against published check values and the guide's own examples
Instrument session, download, plotting, chart, UI	simulator	exercised end to end against a wire-level emulator; no real SWiFT has been connected
Winch depth reporting	stand-in	handshake and dedupe defeat verified against a stand-in for Vigo's listener, not the winch
Unattended download loop	needs hardware	ROADMAP phase 7 — ten consecutive operator-commanded casts, with induced failures
Windows build	outstanding	plat_win32.c is not written, so make windows does not link yet
.vpd reading, batch download, auto-export	not yet	planned, unimplemented
#433 acknowledged extraction	blocked	the protocol figure is blank in our copy of the guide

READ THIS BEFORE QUOTING ANYTHING ABOVE

Simulator-proven is not sea-proven. The emulator was written from the same document as the parsers, so any misreading of the guide is reproduced faithfully on both sides and would go unnoticed until a real instrument is on the bench. What the simulator does buy is that when one is, the remaining unknown is the link and the instrument — not the program.

12 Build and run

```
# Debian/Ubuntu: apt install libsdl2-dev · Arch: pacman -S sdl2 · macOS: brew install sdl2
make                # ./svpview
make test           # every suite, ASan + UBSan
make debug          # sanitised build of the application itself
make windows        # mingw-w64 cross-build (needs plat_win32.c)

./svpview --sim      # run against the emulated instrument
./svpview --open cast.bin # load a logged file
SVPVIEW_SCALE=1 ./svpview # override the detected HiDPI scale
```

Nuklear is vendored in `third_party/`. There is nothing else to fetch, and no network access is required to build or to run.

Where the reasoning lives

File	Contents
ARCHITECTURE.md	layers, threading rationale, crash-resistance, known unknowns
ROADMAP.md	nine phases with what is done and what needs hardware
CHANGELOG.md	the source of truth for release notes, including the bugs found during bring-up
docs/SWIFT_PROTOCOL.md	commands, sentences, the binary header variants
docs/VIGO_INTERFACE.md	the winch messages, the silent-drop rules, and the API not used